PaperMono-Lite

SKU:C153-Lite





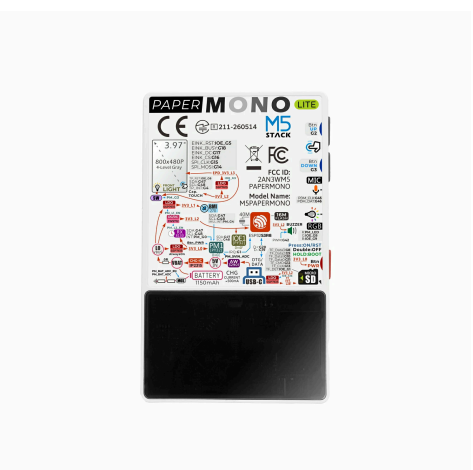


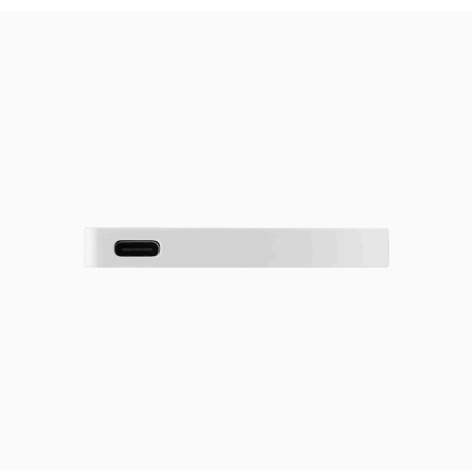














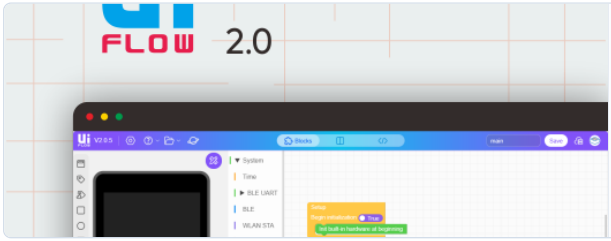




Description

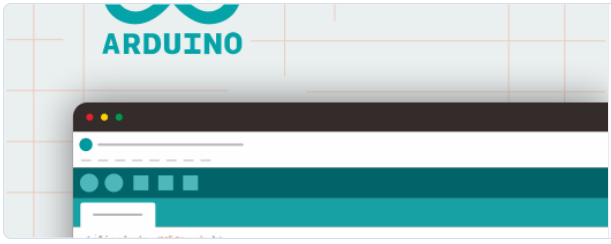
PaperMono-Lite is a streamlined version of PaperMono, designed for low-power display and portable interactive applications. It features a 3.97-inch 4-level grayscale black-and-white E-Ink display with touch support, driven by SSD1677 at 480x800 resolution, and integrates the FT6336G touch controller and a front-light system for clear and stable reading and interaction in low-light environments. The core is powered by the ESP32-S3R8, equipped with 16MB Flash, 8MB PSRAM, and 2.4 GHz Wi-Fi, while also featuring an onboard buzzer, PDM microphone (LMD4737T261-AC02), RGB LED, BMI270 6-axis IMU, M5IOE1 IO expander, microSD card slot, and RX8130CE real-time clock. Compared to PaperMono, the Lite version does not include NFC and LoRa modules, retaining E-Paper interaction and basic wireless connectivity capabilities. With the M5PM1 multi-level power management system and a 1150mAh battery, it is suitable for e-readers, electronic shelf labels, information display terminals, and low-power embedded applications.

Tutorial



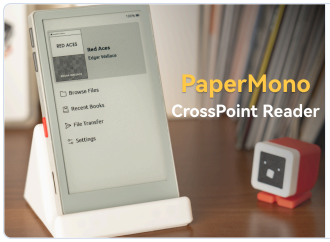
UiFlow2

This tutorial introduces how to control the PaperMono-Lite device using the UiFlow2 graphical programming platform.



Arduino IDE

This tutorial introduces how to program and control the PaperMono-Lite device using Arduino IDE.



CrossPoint E-Reader

Click here to download the PaperMono-Lite CrossPoint E-Reader firmware.

Note

E-Paper Usage Precautions

1. Avoid direct sunlight or prolonged exposure to high temperatures, as heat or strong UV rays may damage the panel.
2. After approximately 10 partial fast refreshes in software, it is recommended to perform a full-screen refresh to clear any accumulated severe ghosting.
3. The screen has built-in OTP waveforms that can be called directly. When using external custom waveforms, pay attention to DC balance; otherwise, irreversible damage to the panel may occur.
4. Avoid continuous partial fast refreshes without interruption to prevent irreversible damage to the panel due to long-term DC imbalance.

Touch Active Area

PaperMono-Lite Touch Active Area

The touch IC firmware internally shrinks the touch boundary. The effective touch coordinate range is limited to: **X-axis: 5 ~ 475** (total width 480px), **Y-axis: 5 ~ 795** (total height 800px).

Features

- ESP32-S3R8 main controller
  - 16MB Flash
  - 8MB PSRAM
  - 2.4 GHz Wi-Fi
- 3.97-inch 4-level grayscale black-and-white E-Ink display with touch support
  - SSD1677
  - 480x800 resolution
  - 4-level grayscale display
  - FT6336G touch
- Integrated front-light system
- Audio interaction
  - PDM microphone (LMD4737T261-AC02)
  - Built-in buzzer
- Onboard peripherals
  - BMI270 6-axis IMU
  - M5IOE1 IO expander
  - microSD card slot
  - RX8130CE RTC
  - RGB LED
- M5PM1 multi-level power management
- Compared to PaperMono, does not include NFC and LoRa modules
- Built-in 1150mAh battery

Includes

- 1 x PaperMono-Lite

Applications

- E-reader
- Electronic shelf label
- Information display terminal
- Portable control panel
- Low-power IoT node

Specifications

Specification	Parameter
SoC	ESP32-S3R8 @ Xtensa® 32-bit LX7 dual-core processor, up to 240MHz
Flash	16MB
PSRAM	8MB Octal
Wi-Fi	2.4 GHz Wi-Fi
Display	3.97" E-Paper (Mono) SSD1677 @ 480x800, with FT6336G touch
Display Grayscale	4-level grayscale display
Front Light	Integrated E-Paper front-light
Input Power	USB Type-C DC 5V
Battery Capacity	1150mAh
Microphone	PDM microphone LMD4737T261-AC02
Buzzer	Built-in buzzer
IMU	BMI270
Power Management	M5PM1
IO Expander	M5IOE1
Expansion Storage	microSD
RTC	RX8130CE
RGB LED	Built-in RGB LED
User Buttons	2x user buttons + 1x power button (ON / OFF / RESET / BOOT)
Product Size	62.0 x 101.0 x 8.0mm
Product Weight	72.4g
Package Size	113.2 x 69.6 x 21.0mm
Gross Weight	88.9g

Learn

Power On/Off

- Power On / Reset: Briefly press the power button once
- Power Off: Press the power button twice in succession

C153-Lite_PaperMono-Lite_poweron_EN.mp4

Download Mode

Press and hold the power button (for approximately 2 seconds) until the red LED flashes, then release it. The device will enter download mode and be ready for firmware flashing.



M5GFX LUT Refresh Speed

The following data are laboratory test results for PaperMono under different M5GFX refresh modes. Actual refresh time may vary depending on display content and operating environment; these values are for reference only.

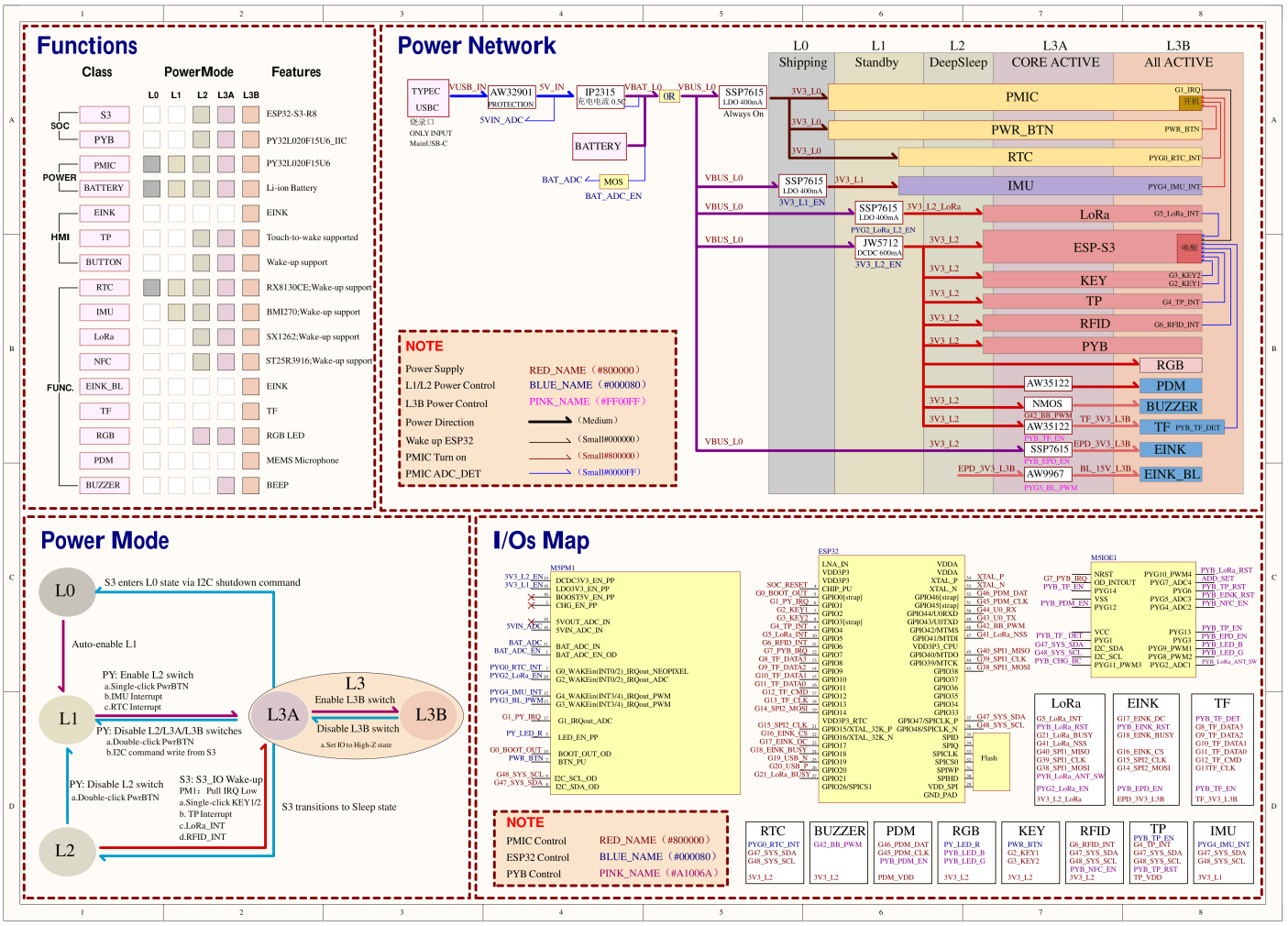
Refresh Mode	Single Refresh Time
`epd_quality`	4.71 s
`epd_text`	0.45 s
`epd_fast`	0.34 s
`epd_fastest`	0.07 s

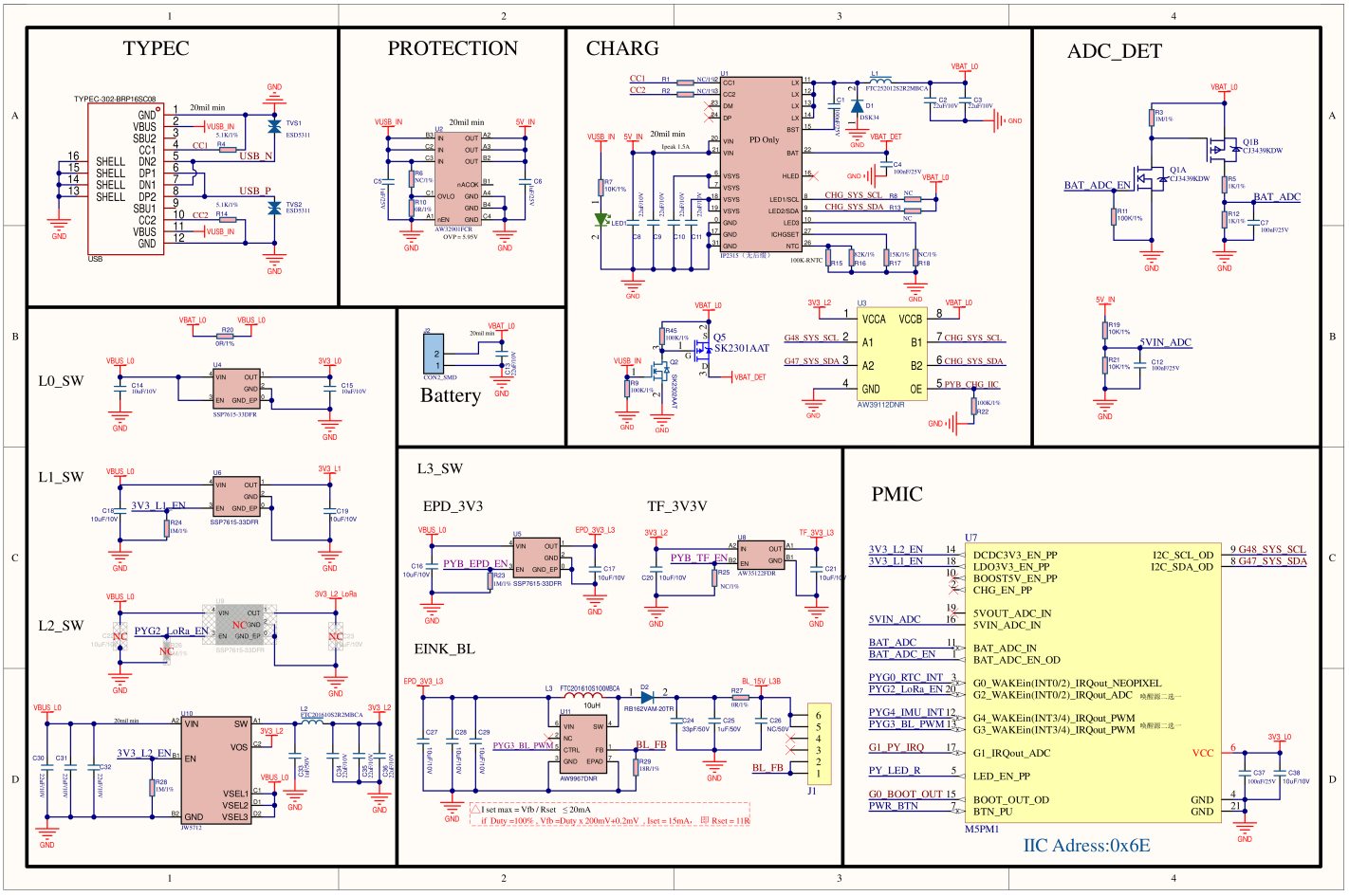
Refresh Mode Demonstration:

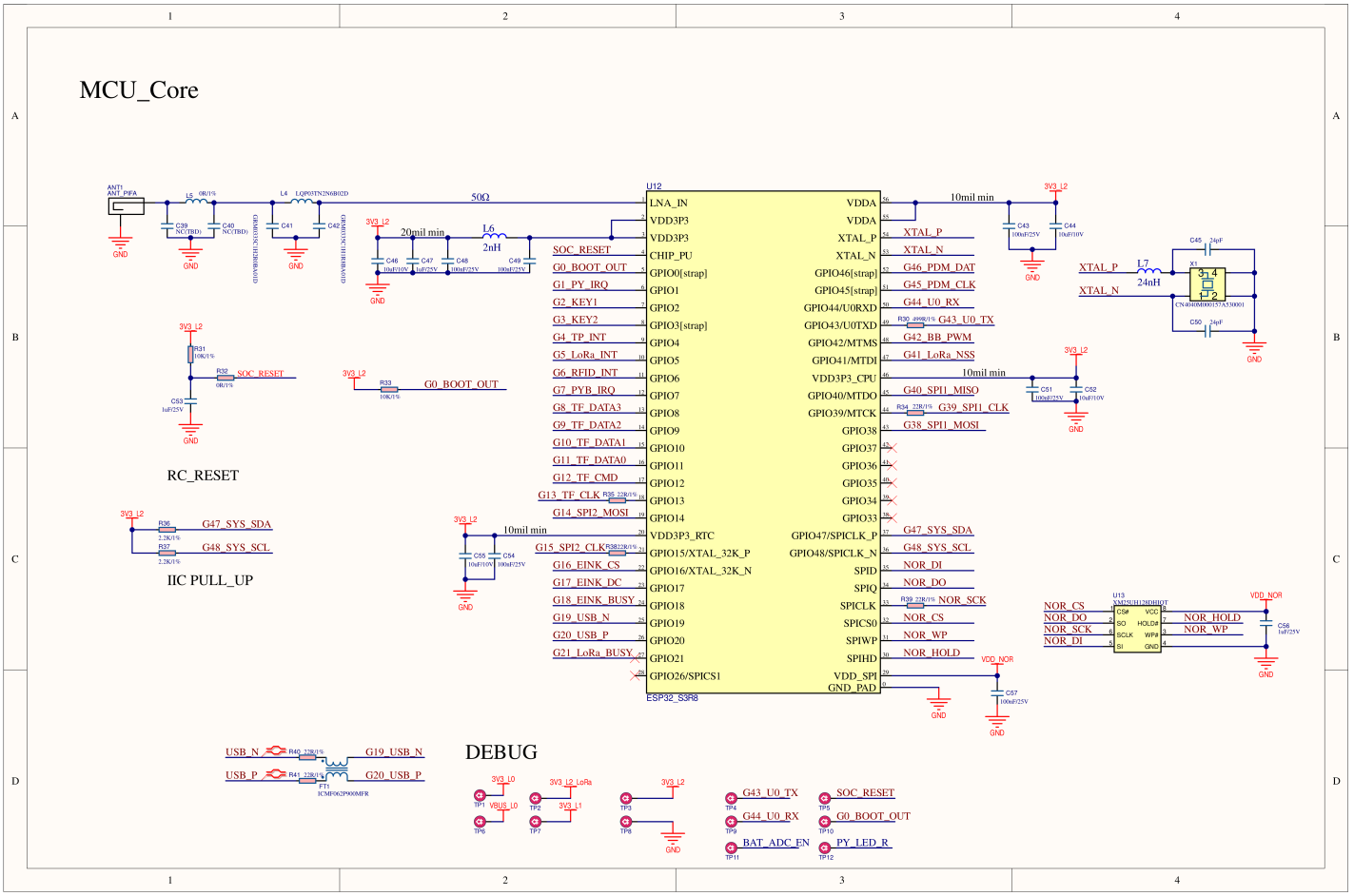
C153_C153-Lite_reflash_mode_demo.mp4

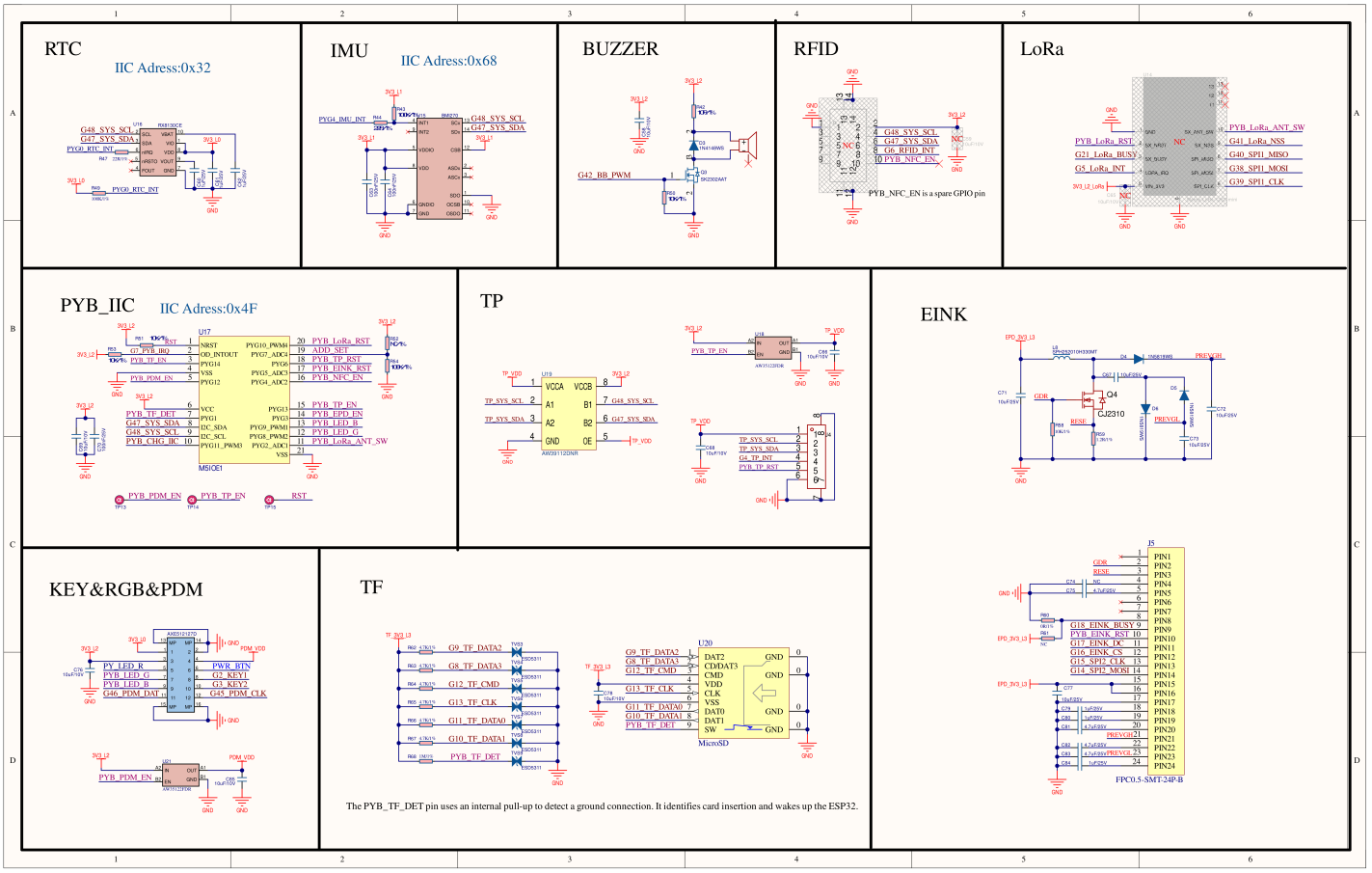
Schematics

- PaperMono-Lite Schematics PDF


Functions
Class
PowerMode
Features
L0 L1 L2 L3A L3B
SOC
S3
PYB
POWER
PMIC
BATTERY
HMI
EINK
TP
BUTTON
FUNC.
RTC
IMU
LoRa
NFC
EINK_BL
TF
RGB
PDM
BUZZER
ESP32-S3-R8
PY32L020F15U6_IIC
PY32L020F15U6
Li-ion Battery
EINK
Touch-to-wake supported
Wake-up support
RX8130CE;Wake-up support
BMI270;Wake-up support
SX1262;Wake-up support
ST25R3916;Wake-up support
EINK
TF
RGB LED
MEMS Microphone
BEEP
Power Network
L0 Shipping
L1 Standby
L2 DeepSleep
L3A CORE ACTIVE
L3B All ACTIVE
TYPEC USBC
AW32901 PROTECTION
IP2315
SSP7615 LDO 400mA Always On
BATTERY
MOS
BAT_ADC_EN
JW5712 DCDC 600mA
PMIC
PWR_BTN
RTC
IMU
LoRa
ESP-S3
KEY
TP
RFID
PYB
RGB
PDM
BUZZER
TF
EINK
EINK_BL
AW35122
NMOS
AW9967
NOTE
Power Supply
L1/L2 Power Control
L3B Power Control
Power Direction
Wake up ESP32
PMIC Turn on
PMIC ADC_DET
RED_NAME（#800000）
BLUE_NAME（#000080）
PINK_NAME（#FF00FF）
（Medium）
（Small#000000）
（Small#800000）
（Small#0000FF）
Power Mode
L0
S3 enters L0 state via I2C shutdown command
Auto-enable L1
L1
PY: Enable L2 switch
a.Single-click PwrBTN
b.IMU Interrupt
c.RTC Interrupt
PY: Disable L2/L3A/L3B switches
a.Double-click PwrBTN
b.I2C command write from S3
L2
PY: Disable L2 switch
a.Double-click PwrBTN
S3: S3_IO Wake-up
PM1: Pull IRQ Low
a.Single-click KEY1/2
b. TP Interrupt
c.LoRa_INT
d.RFID_INT
L3A
L3
Enable L3B switch
Disable L3B switch
a.Set IO to High-Z state
L3B
S3 transitions to Sleep state
I/Os Map
M5PM1
ESP32
M5IOE1
NOTE
PMIC Control
ESP32 Control
PYB Control
RED_NAME（#800000）
BLUE_NAME（#000080）
PINK_NAME（#A1006A）
RTC
BUZZER
PDM
RGB
KEY
RFID
TP
IMU
LoRa
EINK
TF


TYPEC
PROTECTION
CHARG
ADC_DET
L0_SW
L1_SW
L2_SW
Battery
L3_SW
EPD_3V3
TF_3V3V
EINK_BL
PMIC
IIC Adress:0x6E


MCU_Core
ESP32_S3R8
RC_RESET
IIC PULL_UP
DEBUG


RTC
IIC Adress:0x32
IMU
IIC Adress:0x68
BUZZER
RFID
LoRa
PYB_IIC
IIC Adress:0x4F
TP
EINK
KEY&RGB&PDM
TF
The PYB_TF_DET pin uses an internal pull-up to detect a ground connection. It identifies card insertion and wakes up the ESP32.

PinMap

E-Paper

ESP32-S3R8	G14_SPI2_MOSI	G15_SPI2_CLK	G16_EINK_CS	G17_EINK_DC	G18_EINK_BUSY
SSD1677	MOSI	SCLK	CS	DC	BUSY

M5IOE1	PYG5_ADC3	PYG3
SSD1677	RST	EPD_3V3_EN

M5PM1	G3_WAKEin / IRQout / PWM
Frontlight	BL_FB

The E-Paper reset is controlled via `PYG5_ADC3` of `M5IOE1`, and the E-Paper 3.3V power supply is controlled via `PYG3`. The front-light brightness is controlled via `G3_WAKEin` of `M5PM1`.

Touch

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G4_TP_INT
FT6336G(0x38)	SDA	SCL	INT

M5IOE1	PYG6	PYG13
FT6336G	TP_RST	TP_VDD_EN

microSD

ESP32-S3R8	G12	G13	G11	G10	G9	G8
microSD	CMD	CLK	DAT0	DAT1	DAT2	DAT3

M5IOE1	PYG14	PYG1
microSD	EN	DET

`DET` is pulled up by the microSD power domain. When the card slot switch is closed, the detection pin is pulled low to identify microSD card insertion.

HMI

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G2_KEY1	G3_KEY2
RTC - RX8130CE(0x32)	SDA	SCL		
IMU - BMI270(0x68)	SDA	SCL		
KEY			KEY1	KEY2

M5PM1	G0_WAKEin	G4_WAKEin	LED_EN_PP
HMI	RTC_INT	IMU_INT	LED_R

`PYG0_RTC_INT` and `PYG4_IMU_INT` are connected to `M5PM1` and can serve as low-power wake-up sources.

M5IOE1	PYG8_PWM2	PYG9_PWM1
HMI	LED_G	LED_B

The RGB LED indicator on the side of the PaperMono body consists of three color LEDs. The red LED is connected to `LED_EN_PP` of M5PM1 and will light up according to the default behavior of M5PM1 when the device is powered on. Since `LED_EN_PP` does not support PWM output mode, the adjustable color range of this indicator will be limited.

KEY

ESP32-S3R8	G2	G3
KEY	USER_KEY1 (Button A)	USER_KEY2 (Button B)

Audio

ESP32-S3R8	G46_PDM_DAT	G45_PDM_CLK	G42_BB_PWM
PDM MIC	DAT	CLK	
BUZZER			BB_PWM

M5IOE1	PYG12
PDM	PDM_VDD_EN

M5PM1

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G1_PY_IRQ	G0_BOOT_OUT
M5PM1(0x6E)	SDA	SCL	IRQ	BOOT_OUT

M5IOE1

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G7
M5IOE1(0x4F)	SDA	SCL	PYB_IRQ

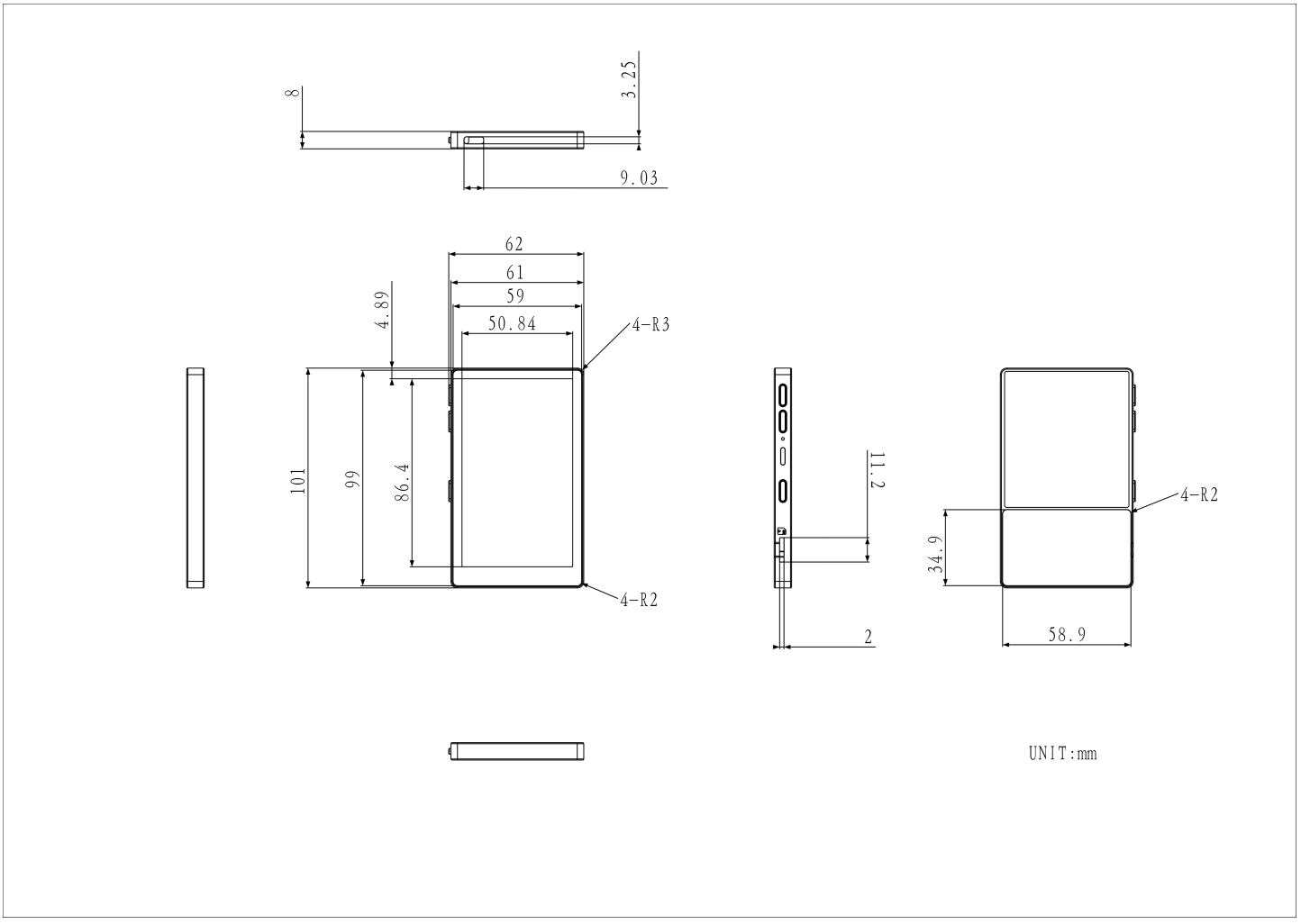
M5IOE1	PYG11_PWM3
Charger - IP2315(0x75)	PYB_CHG_IIC

Note

The I2C operation of the charging chip `IP2315` relies on the I2C pins being pulled up to VBAT voltage. When the device is connected to USB, mode detection is triggered. If VBAT voltage is low, the IP2315 may not initialize to I2C mode properly, which can interfere with other devices on the same I2C bus. M5IOE1 controls the connection of IP2315 to the system I2C bus via PYG11_PWM3. During device operation, it is prohibited to keep the IP2315 permanently connected to the I2C bus; it must be disconnected promptly after communication to avoid reducing bus communication stability. If I2C bus communication anomalies occur, the device can be reset by briefly pressing the power button to restore normal bus operation.

Model Size

- PaperMono-Lite Model Size PDF



Datasheets

- ESP32-S3
- RX8130CE Register Datasheet
- BMI270
- SSD1677 Display Driver
- 3.97-inch Touchscreen

Softwares

E-Paper Driver Notes

1. The e-paper driver waveforms for PaperMono in the M5GFX library are currently unstable. It is recommended to prioritize the e-paper manufacturer's OTP example below for refresh configuration to achieve better panel life and refresh stability.
2. Software recommendation: After approximately 10 partial fast refreshes, perform one full-screen refresh to clear the screen and prevent ghosting from accumulating.
3. After prolonged repeated refreshing, black deposited pixels may appear on the panel due to the physical characteristics of the ink particles. Leave the panel idle for a while, then perform one full-screen refresh to restore it.

Arduino

- PaperMono-Lite Arduino Quick Start
- PaperMono-Lite M5PM1 & M5IOE1 Power Management
- M5PM1 Arduino Library

- M5IOE1 Arduino Library
- M5Unified Arduino Library
- M5GFX Arduino Library

UiFlow2

- PaperMono-Lite UiFlow2 Quick Start

Protocol

- M5PM1 Power Management IC
- M5IOE1 IO Expander IC

ESP-IDF

- PaperMono-Lite Factory Firmware Source Code
- PaperMono-Lite OTP Example
- PaperMono-Lite Low-Power Test Example

PlatformIO

```
[env:m5stack-papermono-lite]
platform = espressif32@6.12.0
board = esp32-s3-devkitm-1
framework = arduino
board_build.partitions = default_16MB.csv
board_upload.flash_size = 16MB
board_upload.maximum_size = 16777216
board_build.arduino.memory_type = qio_opi
build_flags =
    -DESP32S3
    -DBOARD_HAS_PSRAM
    -mfix-esp32-psram-cache-issue
    -DCORE_DEBUG_LEVEL=0
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified = https://github.com/m5stack/M5Unified#develop
    M5PM1 = https://github.com/m5stack/M5PM1
    M5IOE1 = https://github.com/m5stack/M5IOE1
```

Easyloader

Easyloader	Download	Note
PaperMono-Lite User Demo	download	/

Other

- PaperMono-Lite Factory Firmware Restoration
- PaperMono-Lite CrossPoint E-Reader

Video

Product Comparison

Product Compare		
	PaperMono	**PaperMono-Lite**
Main Controller	ESP32-S3R8	ESP32-S3R8
Display	3.97" 4-level grayscale black-and-white E-Paper, 480x800	3.97" 4-level grayscale black-and-white E-Paper, 480x800
Touch	FT6336G	FT6336G
Front Light	Integrated	Integrated
Wi-Fi	2.4 GHz	2.4 GHz
NFC	ST25R3916	❌
LoRa	Stamp LoRa-1262	❌
Expansion Storage	microSD	microSD
IMU	BMI270	BMI270
RTC	RX8130CE	RX8130CE
Battery	1150mAh	1150mAh
Shell Color	Gray	White